Project: Worked examples
Title: Section properties
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Area, centroid, second moments of area (Ixx, Iyy, Ixy), principal axes and values, elastic and plastic section moduli, radii of gyration, torsion constant and (I-sections) warping constant for a parametric shape, an arbitrary polygon with holes, or a composite of placed parts. The x axis is horizontal (major axis of an I-section), y vertical.

Outline loops are integrated by Green's theorem: $A = \tfrac{1}{2}\oint(x\,dy - y\,dx)$, $I_{xx} = \int y^2\,dA$, $I_{yy} = \int x^2\,dA$, $I_{xy} = \int xy\,dA$, transferred to the centroid by the parallel-axis theorem. Principal: $\tan 2\theta = -2I_{xy}/(I_{xx} - I_{yy})$. Elastic modulus I_{xx}/y_{max}; plastic modulus from the equal-area axis found by bisection, $W_{pl} = |Q_{above}| + |Q_{below}|$. Torsion: Saint-Venant series for a solid rectangle, $\pi d^4/32$ for circles, Bredt for closed thin-walled, $\Sigma bt^3/3$ (+ fillet term for rolled I) for open sections. Warping (I): $I_w = I_{yy}(h - t_f)^2/4$.

Ref: Roark's Formulas for Stress and Strain Appendix A, Table A.1; Ch. 10, Table 10.1 — Properties of sections; torsion of solid and thin-walled sections
Ref: AISC Design Guide 9 (Seaburg & Carter) Appendix — Torsional properties of rolled I-shapes
Ref: BS EN 10365 — Hot rolled steel channels, I and H sections: dimensions and masses
Ref: BS EN 10210-2 — Hot finished structural hollow sections: tolerances, dimensions and sectional properties

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
shape	Shape	I-section		spec: BS EN 10365 UB 457×191×98	any
h	Depth h (or vertical leg)	467.2	mm	spec: BS EN 10365 UB 457×191×98	1 … 5000 mm
b	Width b (or horizontal leg)	192.8	mm	spec: BS EN 10365 UB 457×191×98	1 … 5000 mm
t	Wall / web / leg thickness t (t_w)	11.4	mm	spec: BS EN 10365 UB 457×191×98	0.1 … 500 mm
t_f	Flange thickness t_f	19.6	mm	spec: BS EN 10365 UB 457×191×98	0.1 … 500 mm
r	Root radius r (I, channel, tee, angle) or outer corner radius (hollow rectangle; blank = 1.5 t)	10.2	mm	spec: BS EN 10365 UB 457×191×98	0 … 500 mm

3. Assumptions

A1. A polygon outline is simple (no self-intersection); holes lie inside the outline and do not overlap. (Green's theorem integrates the signed loops as given.)
A2. Rolled-section root fillets are quarter circles of radius r; flange tips are square; hollow-section corners are quarter circles of outer radius r_o (default 1.5 t) and inner radius r_i (default 1.0 t). (Published section tables use the same idealisation.)
A3. Thin-walled torsion constants neglect the junction and tip effects except the fillet term kept for rolled I-sections. (Open sections: $\Sigma bt^3/3$; closed: Bredt.)

4. Working

[7] A = ½ ∮ (x dy - y dx) Roark Table A.1
= Green's theorem over 108 vertices in 1 loop(s)
= 125.27 cm²
Area
[8] x_c = ∫x dA / A
= Green's theorem over 108 vertices in 1 loop(s)
= 0 mm
Centroid x (from the shape origin)
[9] y_c = ∫y dA / A
= Green's theorem over 108 vertices in 1 loop(s)
= 0 mm
Centroid y (from the shape origin)
[10] I_xx = ∫y² dA - A y_c²
= Green's theorem over 108 vertices in 1 loop(s)
= 45728 cm⁴
Second moment of area about the centroidal x axis
[11] I_yy = ∫x² dA - A x_c²
= Green's theorem over 108 vertices in 1 loop(s)
= 2347 cm⁴
Second moment of area about the centroidal y axis
[12] I_xy = ∫xy dA - A x_c y_c
= Green's theorem over 108 vertices in 1 loop(s)
= 0 cm⁴
Product of inertia about the centroid
[13] I_1 = (I_xx + I_yy)/2 + sqrt(((I_xx - I_yy)/2)^2 + I_xy^2)
= (45728 cm⁴ + 2347 cm⁴) / 2 + sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)
= 45728 cm⁴
Major principal second moment
[14] I_2 = (I_xx + I_yy)/2 - sqrt(((I_xx - I_yy)/2)^2 + I_xy^2)
= (45728 cm⁴ + 2347 cm⁴) / 2 - sqrt(((45728 cm⁴ - 2347 cm⁴) / 2)^2 + (0 cm⁴)^2)
= 2347 cm⁴
Minor principal second moment
[15] theta = ½ atan2(-2 I_xy, I_xx - I_yy)
= ½ atan2(0, 43381)
= 0 deg
Angle from the x axis to the major principal axis (anticlockwise)
[16] y_top = y_max - y_c
= Green's theorem over 108 vertices in 1 loop(s)
= 233.6 mm
Extreme fibre above the centroid
[17] y_bot = y_c - y_min
= Green's theorem over 108 vertices in 1 loop(s)
= 233.6 mm
Extreme fibre below the centroid
[18] x_ext = max(x_max - x_c, x_c - x_min)
= Green's theorem over 108 vertices in 1 loop(s)
= 96.4 mm
Extreme fibre from the centroid, x
[19] W_el_x = I_xx / max(y_top, y_bot)
= 45728 cm⁴ / max(233.6 mm, 233.6 mm)
= 1957.5 cm³
Elastic section modulus about x (governing fibre)

[20] W_el_y = I_yy / x_ext
= 2347 cm⁴ / 96.4 mm
= 243.47 cm³
Elastic section modulus about y
[21] W_pl_x = |Q_above| + |Q_below| about the equal-area axis
= equal-area axis at y = 0 mm
= 2232.5 cm³
Plastic section modulus about x
[22] W_pl_y = |Q_right| + |Q_left| about the equal-area axis
= equal-area axis at x = 0 mm
= 378.9 cm³
Plastic section modulus about y
[23] i_x = sqrt(I_xx / A)
= sqrt(45728 cm⁴ / 125.27 cm²)
= 19.106 cm
Radius of gyration about x
[24] i_y = sqrt(I_yy / A)
= sqrt(2347 cm⁴ / 125.27 cm²)
= 4.3286 cm
Radius of gyration about y
[25] alpha_pl_x = W_pl_x / W_el_x
= 2232.5 cm³ / 1957.5 cm³
= 1.1404
Shape factor about x
[26] J = (2 b t_f³ + (h - 2 t_f) t_w³)/3 + 2 ⊠ D⁴ - 0.42 t_f⁴
= b = 192.8, t_f = 19.6, h = 467.2, t_w = 11.4, r = 10.2 mm
= 121.29 cm⁴
Torsion constant
[27] I_w = I_yy (h - t_f)² / 4 AISC DG9
= 2347 cm⁴ × (467.2 - 19.6)² mm² / 4
= 1.1755 dm⁶
Warping constant

5. Results

SYMBOL	VALUE	DESCRIPTION
A	125.27 cm²	Area
x_c	0 mm	Centroid x
y_c	0 mm	Centroid y
I_xx	45728 cm⁴	I_xx
I_yy	2347 cm⁴	I_yy
I_xy	0 cm⁴	I_xy
I_1	45728 cm⁴	I_1 (major principal)
I_2	2347 cm⁴	I_2 (minor principal)
theta	0 deg	Principal axis angle
W_el_x	1957.5 cm³	W_el,x
W_el_y	243.47 cm³	W_el,y
W_pl_x	2232.5 cm³	W_pl,x

SYMBOL	VALUE	DESCRIPTION
W_pl_y	378.9 cm³	W_pl,y
i_x	19.106 cm	i_x
i_y	4.3286 cm	i_y
alpha_pl_x	1.1404	Shape factor, x
J	121.29 cm⁴	Torsion constant J
I_w	1.1755 dm⁶	Warping constant I_w

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

WARNING — thin-walled-torsion: J: open thin-walled sum with the rolled-section fillet term.

8. Verification

Template struct.section-properties@1.0.0 Content hash 7f7a93ac64011ba4 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0